

WEB TECHNOLOGIES ASSIGNMENT

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BRANCH:CSM-A

jakarta.servlet Package:

Servlet Interface

Purpose: Core interface for all servlets.

Methods:

1. void init(ServletConfig config) : Initializes the servlet.
2. void service(ServletRequest req, ServletResponse res) : Handles client requests.
3. void destroy() : Cleans up resources before servlet is removed.
4. ServletConfig getServletConfig() : Returns the servlet configuration object.
5. String getServletInfo() : Returns information about the servlet.

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ServletResponse Interface

Purpose: Represents response sent to the client.

Important Methods:

- i. void setContentType(String type) : Sets MIME type of response.
- ii. ServletOutputStream getOutputStream(): Sends binary output.
- iii. PrintWriter getWriter() : Sends character output.

- iv. void setCharacterEncoding(String charset) : Sets response encoding.

ServletConfig Interface

Purpose: Provides servlet initialization information.

Important Methods:

1. String getInitParameter(String name) : Gets initialization parameter.
2. Enumeration<String> getInitParameterNames() : Lists all init parameters.
3. String getServletName() : Returns the servlet's name.
4. ServletContext getServletContext() : Returns servlet context.

ServletContext Interface

Purpose: Represents web application context, allows sharing data across servlets.

Methods:

1. String getInitParameter(String name) : Gets context-wide parameter.
2. Object getAttribute(String name) : Gets attribute stored in context.
3. void setAttribute(String name, Object obj) : Sets attribute in context.
4. RequestDispatcher getRequestDispatcher(String path) : For request forwarding.

GenericServlet Class

Purpose: Abstract class implementing Servlet; allows protocol-independent servlets.

Methods:

1. abstract void service(ServletRequest req, ServletResponse res) : Must be overridden.
2. void init() : Optional initialization.
3. void destroy() : Cleanup method.

ServletException Class

Purpose: Exception thrown by servlets.

Constructors:

- i. ServletException()
- ii. ServletException(String message)
- iii. ServletException(String message, Throwable rootCause)

HttpServletRequest Interface

Purpose: Extends ServletRequest for HTTP requests.

Methods:

- i. String getMethod() : Returns HTTP method (GET, POST, etc.).
- ii. String getRequestURI() : Returns requested URI.
- iii. String getQueryString() : Returns query string.
- iv. Cookie[] getCookies() : Returns cookies sent by client.
- v. HttpSession getSession() : Returns session object.
- vi. String getHeader(String name) : Returns header value.

HttpServletResponse Interface

Purpose: Extends ServletResponse for HTTP responses.

Methods:

- i. void sendRedirect(String location) : Redirects client to new URL.
- ii. void addCookie(Cookie cookie) : Sends cookie to client.
- iii. void setStatus(int sc) : Sets HTTP status code.
- iv. void setHeader(String name, String value) : Sets HTTP header.

HttpSession Interface

Purpose: Represents a user session.

Methods:

- i. String getId() – Returns session ID.
- ii. Object getAttribute(String name) – Gets stored attribute.
- iii. void setAttribute(String name, Object value) – Stores attribute.
- iv. void invalidate() – Ends session.
- v. long getCreationTime() – Returns session creation time.
- vi. long getLastAccessedTime() – Returns last accessed time.

Cookie Class

Purpose: Represents HTTP cookies.

Constructors:

Cookie(String name, String value)

Methods:

- i. String getName() : Returns cookie name.
- ii. String getValue() :Returns cookie value.
- iii. void setMaxAge(int expiry): Sets lifetime.
- iv. void setPath(String uri) :Sets path.